

interpolate_grouped_time_series.R

August 16, 2026

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interpolate_grouped_time_series
Interpolate Grouped Time Series

Description

Interpolates grouped numeric time series to a regular age grid.

Usage

```
interpolate_grouped_time_series(  
  data_time_series = NULL,  
  grouping_variables = "dataset_name",  
  age_variable_name = "age",  
  value_variable_name = "value",  
  interpolation_method = "linear",  
  extrapolation_rule = 1,  
  ties_function = base::mean,  
  age_min = 0,  
  age_max = 12000,  
  time_step = 500,  
  n_cores = 1,  
  show_progress = TRUE  
)
```

Arguments

`data_time_series`
A data frame containing grouping, age, and value columns.

`grouping_variables`
A character vector of column names that define independent time series.

<code>age_variable_name</code>	Name of the numeric age column.
<code>value_variable_name</code>	Name of the numeric value column.
<code>interpolation_method</code>	Interpolation method passed to <code>[stats::approx()]</code> .
<code>extrapolation_rule</code>	Numeric extrapolation rule passed to <code>[stats::approx()]</code> .
<code>ties_function</code>	Function used by <code>[stats::approx()]</code> to combine tied ages.
<code>age_min</code>	Minimum age in the output grid.
<code>age_max</code>	Maximum age in the output grid.
<code>time_step</code>	Positive spacing between consecutive ages in the output grid.
<code>n_cores</code>	Number of workers. Use '1' for sequential ' <code>purrr::map()</code> ' execution or a larger integer for parallel ' <code>furrr::future_map()</code> ' execution.
<code>show_progress</code>	Logical indicating whether supported map operations display progress.

Details

Each unique combination of 'grouping_variables' is interpolated independently with `[stats::approx()]`. Groups that cannot be interpolated return no rows.

Parallel execution temporarily owns the active 'future' plan. The previous plan is restored when the function exits.

Value

A data frame containing the grouping columns and interpolated age and value columns.

See Also

`[stats::approx()]`

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